

Exercise - 9 Solutions

Out-of-Distribution Detection

Exercise 9.1: The OOD Problem

Your CARLA model was trained on sunny daytime images. At deployment, it may encounter images from fog or night - conditions outside its ODD.

1. Why can a standard classifier not be trusted to signal when it receives an OOD input?

→

A standard classifier cannot be trusted to signal OOD inputs because it is trained only to classify inputs into its known classes. It does not learn a separate “unknown” or “outside ODD” class.

Therefore, when it receives foggy or nighttime images, it still produces a normal prediction and confidence score. This is unsafe because standard neural networks can assign high confidence even to OOD inputs.

2. Why is silent failure (confidently wrong prediction) worse than uncertain failure in a safety-critical system?

→

Silent failure is worse than uncertain failure because the system may continue operating normally while being wrong.

For example, if a foggy image contains a pedestrian but the model confidently predicts “no pedestrian,” the planner may not brake.

In contrast, if the model is uncertain, the system can slow down, request human takeover, or switch to a safe fallback mode.

Therefore, in safety-critical systems, detecting uncertainty or OOD inputs is essential to avoid confident but unsafe decisions.

Exercise 9.2: Baseline OOD Detection

Describe the maximum softmax probability (MSP) baseline for OOD detection. How does it work, and what are its main limitations?

→

Maximum Softmax Probability - It is a simple baseline method for detecting out-of-distribution inputs using the classifier's own confidence.

For a classifier, the model outputs probabilities for each known class: $p(y|x)$

$$MSP(x) = \max p(y|x)$$

For OOD detection - the outlier score is defined as

$$D(x) = - \max p(y|x)$$

Case	Maximum softmax probability	OOD score
Confident prediction	High	Low OOD score
Uncertain prediction	Low	High OOD score

Then we choose a threshold: $D(x) > \tau$

Limitations → usually only works to some degree.

Limitation	Explanation
OOD inputs can still get high confidence	May confidently classify foggy, nighttime, or unknown images.
Softmax confidence is not true uncertainty	A high probability does not prove that the input is in-distribution.
Depends on threshold	Choosing the right threshold is difficult and creates a false positive / false negative tradeoff.
Only uses final probabilities	MSP ignores deeper feature information or whether the input actually looks like training data.

Can fail silently	If an OOD image receives high confidence, the system may not trigger a fallback.
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Exercise 9.3: Alternative Methods

Describe one more advanced OOD detection method (e.g., energy score or Mahalanobis distance). How does it improve over MSP?

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Energy Score (Logit based) →

Energy score uses the model's **raw logits** before softmax. Softmax turns these logits into probabilities, but this conversion can make the model look very confident even for OOD inputs.

$$D(\mathbf{x}) = -\log \sum_i^K e^{f_i(\mathbf{x})}$$

Energy value	Interpretation
Low energy	Input likely belongs to the training distribution
High energy	Input may be out-of-distribution

$D(x) > \tau \rightarrow$ If the energy score is above the threshold, the image is treated as OOD.

→ Energy score improves over MSP because -

it uses the **full logit information** (Considers all class logits together) instead of only the maximum probability after softmax.

Mahalanobis Distance Method (Feature Based) →

This method looks at the model's **internal feature representation**.

Idea – When an image input passes through a NN, the model creates feature vectors inside the network. Is this feature vector close to the normal training features, or is it far away?

If it is close to the training distribution, it is likely **in-distribution**.
If it is far away, it may be **out-of-distribution**.

Plot a Gaussian distribution with mean and covariance to in-distribution features, then using Mahalanobis distance to measure how likely a point is under that Gaussian

$$D(x) = \sqrt{(x - \mu_k)^\top \Sigma^{-1} (x - \mu_k)}$$

Distance	Meaning
Small distance	Feature vector is close to normal training features → likely ID
Large distance	Feature vector is far from normal training features → possible OOD

→ Mahalanobis Distance method improves over MSP because -
by checking whether the internal features look similar to the training data but MSP only uses the final confidence score

Note - Mahalanobis does not compare the image to every training image. It compares it to the **estimated distribution** of training features (complete class) unlike KNN - directly compares against stored training examples.

Exercise 9.4: Visualising the Distribution Shift

1. Display five in-distribution (sunny / daytime) images next to five fog and five night images.

→ (ref ipynb for implementation)

2. Also display five images from the different CARLA town. How do they differ from your training images, and how does that difference compare to the fog / night shift?

→ (ref ipynb for implementation)

→ The different-town images are visually closer to the training images than fog or night images because they still have sunny daytime lighting and good visibility. However, they differ in the structure of the environment, such as road geometry, building layout, vegetation, sidewalks, and background objects.

Compared to the fog and night shifts, the different-town shift is more subtle. Fog changes visibility and contrast, while night changes the entire lighting distribution (strong visual shifts)

The different-town images do not strongly change weather or lighting, but they change the scene layout and background structure. Therefore, different-town images are still a distribution shift, but they are less visually obvious than fog and night.

3. Compute the mean softmax confidence of each of your three models on in-distribution vs. fog/night images. Is the model more or less confident on these inputs?

→ (ref ipynb for implementation)

Condition	Pedestrian Detector	Traffic Light Detector	Vehicle Detector
In-distribution	0.9627	0.9884	0.9633
Fog	0.9280	0.8635	0.9078
Night	0.8280	0.9434	0.8548

→

Model	Confidence change
Pedestrian Detector	Confidence drops from 0.9627 to 0.9280 in fog and 0.8280 at night.
Traffic Light Detector	Confidence drops from 0.9884 to 0.8635 in fog, but remains high at night with 0.9434.
Vehicle Detector	Confidence drops from 0.9633 to 0.9078 in fog and 0.8548 at night.

Observation → Even though confidence decreases on OOD data, it is still quite high. Models are still making confident predictions on images outside their training distribution.

Exercise 9.5: Is the Different Town Out-of-Distribution?

The different-town images were taken under nominal weather and lighting - the kind of conditions your system is meant to handle - but in a location your models never saw during training.

1. Look back at the ODD you specified in Exercise 2.2. As written, does your ODD clearly decide whether the different-town images are inside or outside it?

→

- The ODD does not clearly decide whether the different-town images are inside or outside it. The images satisfy the weather, lighting, and camera conditions because they are sunny daytime images with clear visibility and a normal forward-facing camera.
- They also appear to satisfy the scene-type condition because they show urban roads with structured lanes.
- However, the ODD does not explicitly mention whether unseen towns, new road geometries, different building layouts, vegetation, sidewalks, or unfamiliar intersections are allowed.
- Therefore, the different-town images are not clearly outside the ODD like fog or night, but they represent an ambiguous distribution shift.

2. Revise the wording of your ODD so that the new town is unambiguously either inside or outside it. State which you chose and why.

→

Consider it **outside the ODD**.

Revised ODD-

Dimension	Operating / valid	Non-operating / invalid	Runtime detection
Weather	Clear, cloudy, no precipitation	Rain, fog, snow, hail	Image brightness variance drops; dedicated weather classifier
Lighting	Daytime with normal sunlight	Night, dusk/dawn, tunnels, glare	Mean pixel luminance below threshold
Camera	Forward-facing, clean, fixed mount	Obstructed, dirty, tilted, wet lens	Blur score / sharpness metric

Scene type / location	Urban roads with structured lanes only in validated/geofenced CARLA towns used during training and testing	Unseen towns, unseen road layouts, highways, off-road, construction zones	GPS geofence + lane structure detection
Vehicle speed	Low speed / intersection approach, below approx. 50 km/h	High-speed driving	Onboard speedometer reading

→ Choosing this because the different towns may still have sunny daytime weather and structured roads, but its layout was never seen during training. Since road geometry, buildings, vegetation, and intersections can influence the model, the new town represents an unvalidated distribution shift.

In a safety-critical system, it is safer to treat the new town as outside the ODD until the model is tested and validated there.

3. What does that choice imply for how the town images should be treated - as inputs the system must handle correctly, or as inputs an OOD monitor should flag?

→

Because the ODD is revised so that unseen towns are outside the operating domain, the different-town images should be treated as **OOD inputs**.

This means the system is not expected to handle them as normal driving inputs. Instead, an OOD monitor should flag these images as outside the validated operating conditions.

Once flagged, the system should take a safe fallback action, such as reducing speed, requesting human takeover, or disengaging autonomous mode. This is safer than assuming the model will generalize correctly to an unseen town layout.

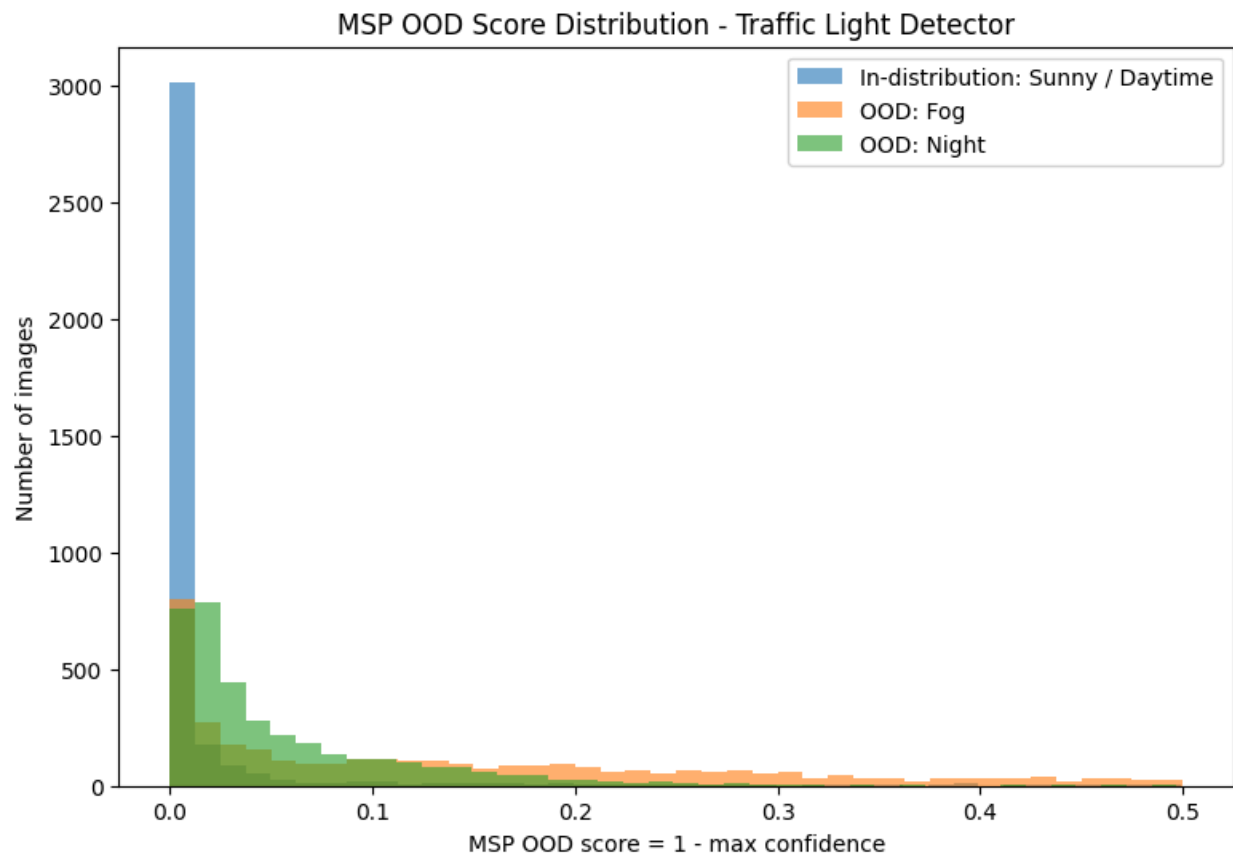
Exercise 9.6: Evaluating the MSP Baseline

Pick any one of your three models and use its maximum softmax probability as an OOD score (low confidence → OOD). Treat the sunny/daytime test images as in-distribution and the provided OOD images as OOD.

1. Plot the distribution of OOD scores for in-distribution and OOD images.

→ (ref ipynb for implementation)

Chosen model - **Traffic Light Detector** (earlier it failed badly at night, so it is interesting for OOD detection)



i) For sunny/daytime in-distribution images, most scores are very close to 0, meaning the model is highly confident.

ii) For fog images, the distribution shifts toward higher OOD scores and becomes more spread out, showing that the model is less confident under foggy conditions.

iii) For night images, the distribution is also more spread out than the sunny distribution, but many night images still receive low OOD scores. This means the model remains confident on many night inputs even though they are outside the ODD.

Overall, MSP detects some distribution shift, especially for fog, but it is not fully reliable because the OOD and in-distribution score distributions overlap. Hence, MSP is not very useful here.

2. Compute the AUROC for separating in-distribution from OOD, over all of the OOD scenarios.

→ (ref ipynb for implementation)

AUROC - The probability that a randomly chosen OOD image receives a higher OOD score than a randomly chosen in-distribution image.

In this case **AUROC = 0.8858**

→ The AUROC for separating sunny/daytime in-distribution images from fog and night OOD images using the MSP OOD score is **0.8858**. This indicates that the MSP baseline provides reasonably good separation between ID and OOD inputs.

An OOD image receives a higher OOD score than an in-distribution image in about **88.58%** of random pairs.

As per the plot, the score distributions still overlap, meaning that some fog or night images are assigned low OOD scores and are therefore treated as confident in-distribution predictions. This confirms that MSP is useful as a simple baseline, but it is not reliable enough as the only safety mechanism for OOD detection.

→ Individual AUROC's with Fog and Night

Fog AUROC after MSP computation : **0.8894**

Night AUROC after MSP computation : **0.8822**

Exercise 9.7: Feature-Based OOD Detection

Implement one feature-based detector (Mahalanobis distance or k-NN, both covered in the lecture) and compare it to the MSP baseline.

1. Extract deep features for your in-distribution training (or validation) images and for the test in-distribution and OOD images. Use the same model whose MSP you evaluated in Exercise 9.6.

→

(ref ipynb for implementation)

2. Fit the detector on the in-distribution features only, then score the test images (larger distance → more OOD).

Used **k=5**,

If k=1, the OOD score depends only on the nearest training feature, which can be noisy and may fail if an OOD image is accidentally close to one in-distribution example. If k is too large, the distance is averaged over many neighbours and may become too smooth, reducing the ability to detect local distribution shifts. Therefore, k=5 is providing a balance between sensitivity and stability.

In a real safety-critical setting, k should be tuned on validation data and evaluated across different OOD scenarios.

Using this formula -

$$\text{kNN OOD score}(x) = \frac{1}{k} \sum_{i=1}^k d(x, N N_i)$$

KNN Scores -

Dataset	Mean k-NN score	Significance
ID sunny/daytime	4.6457	Close to training feature distribution
Fog	9.1769	Much farther from ID features
Night	9.7064	Farthest from ID features

→ The sunny/daytime test images are close to the in-distribution feature space, while fog and night images are far away from it. Night images are the most different from the in-distribution feature space.

3. Compute its AUROC and compare it to the MSP AUROC from Exercise 9.6. For which OOD scenario is the gap largest?

→

The largest gap occurs for **night images**, where the AUROC improves by **0.1121**. This means the feature-based k-NN detector is especially better at detecting night images as OOD. The reason is that MSP only uses final confidence, while k-NN checks whether the image's deep feature representation is far from the in-distribution feature space.

OOD scenario	MSP AUROC	k-NN AUROC	Difference
Fog + Night combined	0.8858	0.9880	+0.1022
Fog only	0.8894	0.9816	+0.0922
Night only	0.8822	0.9943	+0.1121

→ k-NN separates in-distribution sunny/daytime images from OOD images much more reliably than MSP.

Exercise 9.8: Extending the Safety Analysis for OOD

You have now seen how easily the perception models fail - silently - on out-of-ODD inputs, and how well (or poorly) an OOD monitor can catch them. Revisit your System-Theoretic Process Analysis from Sheet 2 and extend it to cover this risk. Many of you will not yet have an OOD-related entry in your analysis; the goal of this exercise is to add one and trace it through.

1. Hazard. Check the hazards from Exercise 2.4. Is operating on an undetected outof-ODD input captured (e.g. as a cause of “vehicle fails to brake for a pedestrian”)? If your hazard list does not reflect it, refine or add a hazard.

→

ID	Hazard	Loss(es)	Likelihood	Severity
H-1	Pedestrian in vehicle path while vehicle is moving without braking	L-1, L-2	Medium	High
H-2	Vehicle continues through intersection with undetected traffic signal	L-3, L-4	Medium	High
H-3	Vehicle fails to detect oncoming/crossing vehicle and does not yield	L-2, L-3	Medium	High
H-4	Autonomous mode remains active during ODD violation, e.g. night, rain	L-1, L-2, L-3	High	High

H-5	Vehicle brakes unnecessarily at high frequency due to false positives	L-2, L-3	Medium	Low
H-6	Vehicle continues operating on an undetected out-of-ODD input, such as fog, night, or unseen town layout, while perception models give confident but unreliable predictions	L-1, L-2, L-3, L-4, L-5	High	High

→ Operating on an undetected out-of-ODD input was only partially captured in the original hazard list. It was related to **H-4: autonomous mode remains active during an ODD violation**, but the hazard list did not explicitly describe silent perception failure on OOD images.

2. Unsafe control action. Review the UCAs from Exercise 2.6. Add a UCA for the case where the planner acts on perception output that is unreliable because the input is out-of-distribution and this is not detected (e.g. “the planner continues at speed while the camera input is out-of-ODD and the perception output is untrustworthy”). Link it to the hazard(s) above.

→

UCA ID	Controller	Control action	Unsafe control action	Linked hazard(s)
UCA-13	Planning module	Continue driving / maintain speed	The planner continues driving at normal speed while the camera input is out-of-ODD and the perception output is unreliable, because the OOD condition is not detected.	H-1, H-2, H-3, H-4, H-6

→ This UCA is linked to **H-6**, where the vehicle continues operating on an undetected out-of-ODD input while the perception model gives confident but unreliable predictions. It is also linked to **H-1, H-2, and H-3**, because unreliable perception can cause the planner to miss pedestrians, traffic lights, or crossing vehicles. It is additionally related to **H-4**, because the autonomous mode remains active during an ODD violation instead of switching to a safe fallback mode.

3. Safety constraints. From that UCA, derive at least one new safety constraint of each kind (cf. Exercise 2.7):

- a model-level constraint on the OOD monitor, justified by the severity of the hazard it guards against; and
- a system-level constraint on the response.

Constraint ID	Type	Safety constraint	Justification
SC-13	Model-level	The OOD monitor shall detect out-of-ODD camera inputs such as fog, night, or unseen town layouts with sufficiently high recall before the planner uses the perception output for driving decisions.	This guards against H-6, where the vehicle continues operating on undetected OOD input. Because this can lead to missed pedestrians, missed traffic lights, or missed vehicles, the severity is high. Therefore, the OOD monitor should prioritize high recall, even if this creates some false alarms.
SC-14	System-level	If the OOD monitor flags an input as out-of-ODD or uncertain, the system shall not continue normal autonomous driving. It shall reduce speed to a safe limit, increase caution, and request takeover or disengage autonomous mode.	This prevents UCA-13 by ensuring that unreliable perception output is not blindly used by the planner. Even if the classifier gives a confident prediction, the system must switch to a safe fallback when the input is outside the validated ODD.

4. Residual risk. Even with a perfect OOD detector, what risk remains?
Does detecting OOD fully address the UCA and hazard?

→

Even with a perfect OOD detector, some risk still remains.

Detecting OOD helps prevent the system from **silently trusting unreliable perception output**, but it does not fully solve the hazard by itself. The system still needs to respond safely after OOD is detected.

Further risks →

Residual risk	Explanation
Safe fallback may fail	The system may detect OOD but fail to slow down, stop, or request takeover correctly.
Too little time to react	OOD may be detected late, for example when the vehicle is already close to a pedestrian or intersection. Distance needs to be considered as an important feature.
Human takeover may not happen quickly	If the system requests takeover, the human may be distracted or unable to respond in time.
OOD detector scope is limited	It may detect fog/night, but not all possible unsafe conditions, sensor faults, adversarial triggers, or unusual objects.
In-distribution failures still exist	Even inside the ODD, the perception model can still miss pedestrians, traffic lights, or vehicles.
False positives can reduce usability	If the detector flags too often, the system may brake unnecessarily or disengage frequently.

→ OOD detection only addresses one part of the problem.

A perfect OOD detector can identify the OOD condition, but the system must still take a safe control action. If the planner ignores the OOD signal or responds too late, the hazard can still occur.

OOD detection does not address all perception failures. The model may still fail on in-distribution images, such as missing a pedestrian in normal sunny daytime conditions. Other risks also remain, such as sensor faults, delayed braking, human takeover failure, and unnecessary braking due to false OOD alarms.

Therefore, OOD detection reduces the risk of silent failure, but it does not fully eliminate the UCA or hazard. It must be combined with system-level fallback behaviour, such as slowing down, requesting takeover, disengaging autonomous mode, or performing a minimum-risk maneuver.